



University of Science and Technology of Hanoi · Department of Space and Earth Sciences

M1 Internship Defense · hosted by QTA Lab, PTIT

Weather-Aware Scheduling for Simultaneous Information and Key Distribution over Tropical Satellite Free-Space Optical Channels

Student: Truong Tuan Nghia (2540017), Master Space Program (M1)

External Supervisor: Dr. Vu Quang Minh, QTA Lab, PTIT

Internal Supervisor: Dr. Phan Thanh Hien, USTH

Hanoi, August 2026

Motivation: a network problem hiding inside a physics problem

- Quantum key distribution (QKD) lets two parties share a secret key whose secrecy follows from **physics, not computational hardness** – unbreakable even by a future quantum computer.
- From orbit, QKD reaches continental distances with far less loss than fiber; the Micius satellite proved this in 2017.
- But over Southeast Asia, cloud does not just weaken the optical link – it **blocks it outright** for much of the year, and an operator manages *several* ground stations and a live megaconstellation, not one clean link.

Question. What is a tropical satellite quantum network actually worth, once real weather, real orbits, and multiple competing ground stations are all accounted for at once?

System concept and the three questions this study answers

SIKD: one 1550 nm carrier, two channels via subcarrier multiplexing – classical data and quantum key, sharing one photodetector (so the data channel leaks into the key channel as cross-talk noise).

Q1. How often is a tropical link usable, under real climatology?

Q2. Does weather-aware scheduling help, judged *fairly*, at equal network resources?

Q3. If the gain is small, is the bottleneck the algorithm – or the network itself?



(a) One SIKD link. (b) The network problem this study solves.

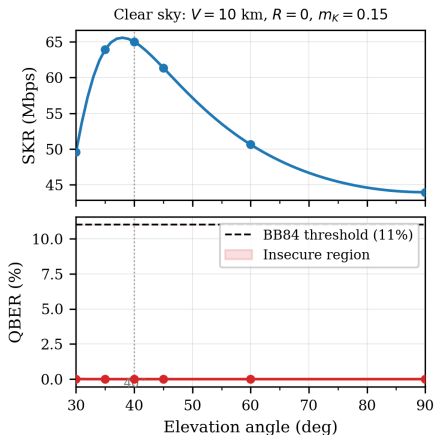
Physical layer: channel model and a real ten-year weather record

Channel factorizes as $h = h_g \cdot h_l \cdot h_a$:
geometric spreading (over 30 dB alone),
atmospheric attenuation, weak log-normal
turbulence.

Weather is a **three-state model** per
city/month – clear, rain, cloud-blocked –
calibrated on ten years (2015–2024) of real
ERA5/ERA5-Land reanalysis:

$$\text{SKR}_{\text{eff}} = p_{\text{clear}} \text{SKR}_c + p_{\text{rain}} \text{SKR}_r$$

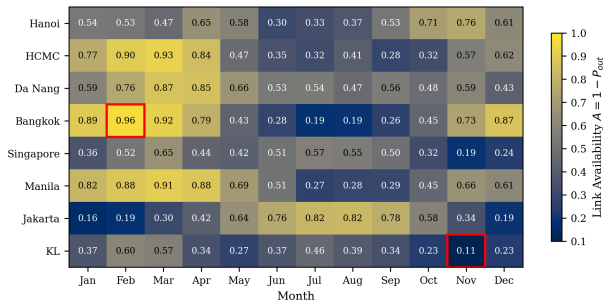
At 45°: $\text{SKR} \approx 61.4 \text{ Mbps}$, $\text{QBER} < 0.001\%$,
well under the 11% BB84 threshold across
30°–90°.



SIKD performance vs. elevation, clear sky.

Answering Q1: link availability across ASEAN

- Availability ranges from **10.7%** (Kuala Lumpur, November) to **96.5%** (Bangkok, February).
- Driven almost entirely by **cloud outage**, not rain attenuation.
- A clear-sky key rate – the number most prior work reports – therefore describes only a *minority* of real operating hours.



Link availability, 8 ASEAN cities across 12 months,
10-year ERA5 climatology.

Real orbits: coverage geometry and city-pair feasibility

- Orbits propagated with SGP4 from **live** Shell-1 two-line elements, 1,019 real Starlink satellites.
- Geometry alone splits the 28 ASEAN city pairs: **14 DUAL** (share one satellite pass) vs. **14 store-and-forward** (relay through a trusted node).
- This split is a *ceiling* fixed by geography – no scheduler can lift it.

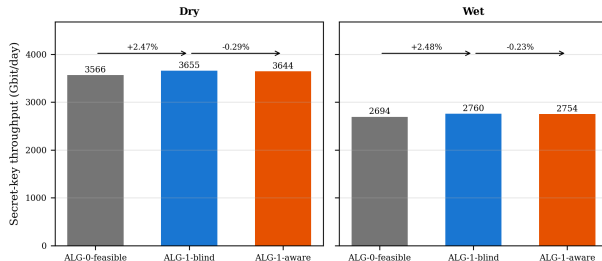
Hanoi	--	DUAL	DUAL	DUAL	SF	SF	SF	SF
HCMC	DUAL	--	DUAL	DUAL	DUAL	SF	SF	DUAL
Da Nang	DUAL	DUAL	--	DUAL	SF	DUAL	SF	SF
Bangkok	DUAL	DUAL	DUAL	--	DUAL	SF	SF	DUAL
Singapore	SF	DUAL	SF	DUAL	--	SF	DUAL	DUAL
Manila	SF	SF	DUAL	SF	SF	--	SF	SF
Jakarta	SF	SF	SF	SF	DUAL	SF	--	DUAL
Kuala Lumpur	SF	DUAL	SF	DUAL	DUAL	SF	DUAL	--
	Hanoi	HCMC	Da Nang	Bangkok	Singapore	Manila	Jakarta	Kuala Lumpur

■ DUAL (simultaneous downlink) ■ SF (store-and-forward)

City-pair feasibility: green = DUAL, orange = store-and-forward.

Answering Q2: a fair, equal-resource comparison

Two levels: (1) match stations to satellites – solved *exactly* by the Hungarian algorithm; (2) allocate each pass's key to the city pair that needs it. Three strategies at **equal resources**, every handover charged its real 30-second cost, over **620 real days**. **Result.** Coordination gains **+2.47%** (dry) / **+2.48%** (wet); weather-awareness on top **costs -0.29%** / **-0.23%** – it re-assigns more often, and handovers outweigh the benefit.

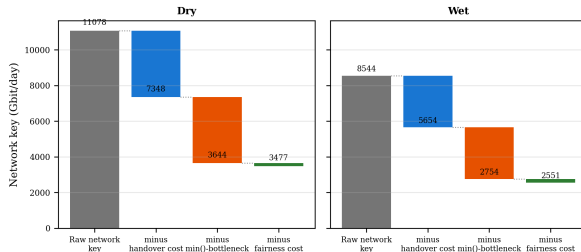


Fair pair-key comparison, full 620-day enumeration.

Answering Q3: the bottleneck is structural, not algorithmic

- On **91.6–96.5%** of days, at least one of 28 city pairs receives **zero** usable key – regardless of algorithm.
- Cause: 8 stations cannot serve 28 pairs. This is a capacity-planning finding, not a scheduling failure.

Open tension. The climatologically best window is mid-morning – in daylight – while every reported SKR assumes night-time operation with zero background light.



Where delivered key is lost along the pair-key path.

Conclusions and what comes next

Contributions

1. A complete channel-to-key chain, evaluated on real weather (10-yr ERA5) and real orbital geometry (live TLEs) – not idealized inputs.
2. A fair, equal-resource comparison of three scheduling strategies across 620 enumerated real days, with statistical significance testing.
3. Two structural findings with direct engineering value: pair-starvation is a capacity-planning limit, and weather-awareness costs handovers before it saves anything.

Central claim. Tropical satellite quantum networks should be judged by fairly compared, network-level scheduling outcomes – not an idealized single-link key rate under clear skies cities rarely get. **Next.** Daytime channel validation, station/HAP siting optimization, and a composable security proof for the trusted-relay model.